MATH 213 Statistical Concepts and Methods

Spring 2025 *Updated: 1/21/25*

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Semester at a glance

Prerequisite

MATH 120 (Precalculus) or Level III placement on the Basic Math Skills Inventory

Textbook

Introduction to Modern Statistics by Mine Cetinkaya-Rundel and Johanna Hardin. Electronic copy of the textbook here: https://openintro-ims.netlify.app/. You can buy a PDF copy of the textbook at a "pay what you can" price there, or you can buy a paperback copy of the textbook from Amazon.

Grading

I will calculate your course grade based on the assignments described below with the following weights:

Quizzes	10%
Homework and graded classwork	20%
Projects	15%
Two midterms during the semester	30%
Comprehensive final exam	25%

After computing your weighted average, I will round your grade to the nearest whole number and assign your letter grade according to the grade breakdown in the Hood College course catalog.

Important (tentative) dates

Midterm 1: Friday, February 21st

Midterm 2: Friday, April 4th

Final Exam: Friday, May 9th, 8:30-11am

Midterm Project: coming soon

Final Project: coming soon

Course Description and Purpose

Course description

From the Course Catalog: *An introduction to the practice of statistics, its applications, and its mathematical underpinnings. Focus is on data, activities, technology, conceptual understanding.* (4 credits)

A note from me: First and foremost, this is a statistics class, NOT a math class. Mathematics will be used as a tool to help us with our primary work of storytelling through data. We will learn to visualize and summarize data using R, understand populations using sample data, draw conclusions from a randomized experiment, and communicate our findings in writing. In this course, you will find that we will focus much more on writing than computation, which may be surprising to you given the class has a MATH prefix. In short: the computations and graphs we create are worthless if we can't explain their meaning.

Goals

There are five main goals for this course to which all the course content goals can be aligned. I hope that by the end of the semester, you will…

1. Be critical consumers of statistical content. You will be able to make sense of and hold a healthy amount of skepticism for statistical information from your everyday lives and in academic papers.
2. Be able to identify research questions and state which types of analyses are appropriate for a given dataset.
3. Be able to use the statistical computing software, R, to manipulate data, present it graphically, and summarize it numerically.
4. Be able to draw carefully qualified conclusions from your statistical analysis of data.
5. Be proficient in communicating the findings of your analyses.

Quantitative literacy

This course fulfills the Quantitative Literacy (QL) component of Hood's core curriculum. Hood's learning outcomes for all such sources state that at the end of such a course, you should be able to:

1. Interpret quantitative data arising in a variety of contexts. (Big Picture (BP) goals 1, 3, 4, 5)
2. Demonstrate computational fluency, including the use of technology as appropriate. (BP goal 3)
3. Create and communicate arguments using quantitative tools such as tables, graphs, and mathematical expressions. (BP goals 3, 4, 5)
4. Create and communicate arguments through narrative analysis. (BP goals 3, 4, 5)

Alignment with outcomes for the mathematics major

This course also supports two overall goals for the mathematics major:

1. Students will demonstrate skill in written communication of mathematics through writing about mathematics coherently and logically with mastery of disciplinary style and conventions. (BP goal 4, 5)
2. Students will make effective use of the computer as a tool in mathematics through using a statistical package to perform calculations and interpreting the results. (BP goal 2, 3, 4)

Course Materials

Blackboard

Blackboard will be the source of all information pertaining to our course this semester, including in-class materials and at-home assignments. Please consult our course site regularly. I will also communicate with our class by emailing you through Blackboard. Set up two-factor authentication in a way that's convenient for you and doesn't require cell reception**. I strongly recommend using the Microsoft Authenticator app**, because sometimes our classrooms have bad cell reception.

Software

We will use the statistical computing language R this semester. We will do this via Posit Cloud (formerly RStudio Cloud), which is a cloud-based environment that enables us to use R from anywhere using the internet. Your access to Posit Cloud has been paid for by the Department of Mathematics for this semester.

Assignments and grading

Homework

I will assign homework regularly and you should do it. You should view the homework as preparation for the quizzes, projects, and exams. It also gives you the opportunity to self-assess your understanding so that you are equipped to ask clarifying questions the next day in class or during office hours.

I'll post assignments on Blackboard. It is your responsibility to check Blackboard to find homework assignments and to keep track of deadlines.

Homework should be submitted on Blackboard by the beginning of class the day it is due to count as being "on time". However, following our in-class discussion of the homework, you may make revisions to your assignment if you wish and resubmit at the same Blackboard link. Such revisions are due no later than 11:59pm the day the assignment is due. I strongly encourage you to take advantage of this discussion/revision process.

You are also responsible for checking your work carefully against the posted homework solutions. If I leave feedback on your work, you should read it and incorporate that feedback on your next assignment. I am happy to answer any questions you might have about the homework, either in class (time-permitting) or during office hours.

I will drop your two lowest homework scores at the end of the semester in the calculation of your overall grade.

Class work

I'll ask you to occasionally submit your class work for a grade. I'll announce that in class and on Blackboard.

R Labs

We will work on R lab assignments this semester. Some of the work we do in R will be in-class and ungraded, as part of a formative activity. Some of the work we do in R will be graded and will count in the Homework and Graded Classwork part of your grade.

Quantitative Literacy (QL) assessment

Since this course satisfies the QL part of Hood's Core Curriculum, I will need to assess your learning on the quantitative literacy goals stated earlier in this document. This assessment will be a take-home assignment that will be administered approximately half to two-thirds of the way through the semester. This assignment will be worth twice as much as a usual homework assignment and will count towards the Homework part of your grade.

Quizzes

On most Fridays, we will have in-class group quizzes. You will be assigned to a team of 2-4 students and will work collaboratively to answer questions based on material we have recently covered in class. I will drop your lowest quiz score in the calculation of your overall grade at the end of the semester. You must generally be in class on the day the quiz is held in order to take the quiz. If you have an excused absence on a quiz day, you may email me ahead of time to take the quiz at another time.

Projects

There will be two projects this semester, a midterm project and a final project. Both will involve using R to analyze a dataset and writing a report. The midterm project will have you analyze a specific dataset. The final project will ask you to analyze a dataset that you will choose in consultation with me. More details will be provided as the semester progresses.

Exams

We will have two midterm exams and a cumulative final exam. All exams will be held in our usual classroom. These dates may be revised based on our progress in the course. If they need to be changed, I will announce the change in class and on Blackboard.

Late work

Since I drop two homework assignments and one quiz, I am fairly strict about late work:

Homework should be submitted on time so that you don't fall behind. However, I will accept homework late, up until the day that it is "returned" to the class, albeit with a late penalty of one letter grade per class period it is late.

If you have an excused absence on a quiz day, you can contact me **ahead of time** to take the quiz at another time.

Late exams will only be given under extreme circumstances, and I'll need official documentation of the absence from the Dean's office.

Attendance and participation expectations

Your attendance and participation are expected. I expect that you will actively engage with whatever activities we have for that class meeting. You should give them your best shot and participate with your classmates. I do my best to post these activities ahead of time so you can look at them in advance of the class if you want.

If you must miss a class, it is your responsibility to find out what you missed that day. You should complete the in-class activities on your own and reach out to a classmate for additional notes. If, after you have done all that, you have questions, feel free to reach out to me with those questions. ***It is not appropriate to request a one-on-one meeting with me to discuss missed material without having worked on it yourself.***

I may excuse your absence in the case of documented/authorized Hood college activities, or in other extremely limited cases. If you believe your absence should be excused, it is your responsibility to notify me ahead of time. In the case of family emergencies, extended illnesses, or similar, you should contact the Dean of Students office (deanofstudents@hood.edu) and they will notify me that your absence is excused.

For the occasional unexcused absence due to illness, travel, etc, you don't need to contact me unless you have questions about the day's work. If you have more than five unexcused absences, I'll contact the registrar's office about withdrawing you from the class with a grade of WX.

Academic integrity

Academic Honor Code and the Code of Conduct

As a place of honor and respect, all members of Hood College assume the obligation to maintain the principles of honesty, responsibility, and intellectual integrity in all activities related to their Hood College experience. Students are expected to adhere to the highest standards of academic honesty and integrity in all coursework and related matters. It is the responsibility of each student to support these values through maturity of thought, expression, and action. Members of the faculty and staff are available to assist students in this process. The Academic Judicial Council, which is comprised of undergraduate students and two elected faculty members, is chaired by Dean Paige Eager. If you have questions about the Academic Judicial Council or the Hood College Honor Code, please contact Dean Eager at eager@hood.edu. A complete description of Hood's Academic Honor code is available in Appendix B of the Student Handbook which is available at http://www.hood.edu/policies/ . This page also contains information about Hood's Code of Conduct as well as information about Title IX (see below) and the student conduct process.

Generally, I expect that the work you submit will be your own. You may work with your classmates, but you should write your answers in your own words.

Here are some guidelines for authorized aid on various types of assignments. If you are ever uncertain about whether a certain form of aid is allowed, please ASK before using it. **Students using unauthorized sources of aid will earn a zero as a minimum penalty**. We will continue to discuss this as the semester progresses.

Written homework	Work in R	Projects	Exams	Quizzes
✔ Your textbook, our materials on Blackboard, the notes you took in class. You may collaborate with your classmates on homework, but you should write up your own homework solutions.	✔ Your textbook, our materials on Blackboard, the notes you took in class. Cited sources from other textbooks, the internet, or AI (see further guidance below). You may collaborate with your classmates on work in R via discussion, but you must type up your own work.	✔ Your textbook, our materials on Blackboard, the notes you took in class. **For R Code ONLY:** Cited sources from other textbooks, the internet, or AI.	✔ A two-sided 8x11in page of notes that must be created by you.	✔ Your textbook, our materials on Blackboard, the notes you took in class, each other.

Use of online resources[1]

There is quite a lot of helpful R code available on the internet (e.g. on StackOverflow.com). The policy for this class is that you may use code found on the internet, modified for your own purposes, as long as you cite the source. To use code from another source without an appropriate citation will be considered plagiarism.

Use of Generative AI Tools

You should treat generative AI, such as ChatGPT, the same as other online resources. There are two guiding principles that govern how you can use AI in this course: (1) Cognitive dimension: Working with AI should not reduce your ability to think clearly. If used, AI should facilitate rather than hinder learning. (2) Ethical dimension: Students using AI should be transparent about its use and make sure that it aligns with academic integrity.

[1] This section of the syllabus on use of online resources including generative AI is a modified version of the analogous section from the syllabus of Mine Cetinkaya-Rundel, https://mine-cr.com/teaching/sta101/.

✓**AI tools for code:** You may make use of the technology for coding examples on assignments. If you do so, you must explicitly cite where you obtained the code. You can use [these guidelines]() for citing AI use.

✗ **AI tools for narrative:** Unless instructed otherwise, you may not use generative AI to write narrative on assignments, including short-answer responses to textbook questions and longer, written passages. The work you turn in should be your own and it should accurately reflect your understanding of the course content.

Phone and technology policy

To help keep us focused, please keep your use of technology for non-class-related purposes during class to a bare minimum. This includes excessive texting (or similar), checking social media, playing video games, watching videos, sending email, etc. For students who benefit from some extra analog stimulus during class, I recommend (and encourage) doodling, sudokus, crossword puzzles, recreational reading, arts and crafts, or literally anything else that doesn't involve a screen.

Inclement weather

If Hood closes in-person classes due to weather, I will email you to tell you if we'll be able to hold a Zoom class or if I'll have to cancel.

Where to go for help

Me

If you have questions about the homework, about your grade, or about anything else to do with our class, please feel free to see me in my office hours. If you can't see me during my office hours, email me and we'll work out a time to meet. I will bend over backwards to help you understand.

Each other

One of many reasons I require working with your classmates in class is to build your support network outside of class. I strongly encourage working with each other on homework and to study for exams.

Tutoring services

There are two free tutoring options available to you this semester. The first is peer-tutoring through the College's Student Success Center. The second is free, professional tutoring through the online contracted service ThinkingStorm. See the "Tutoring" link in Blackboard for more details about both these options.

A Warning about searching for online help

There are millions of pages and videos offering statistics help, and I'm sure many of them are great. But I'll warn you that many of these pages and videos talk about statistics in a different way than we do. They may go into a lot of detail about formulas and other things that aren't important for our understanding but skip other aspects that I think are vital. I wouldn't prevent you from searching for more information online even if I could, but please be aware that it may be a real waste of your time. Talking to me will almost certainly be faster.

Tech support

For help from Hood's IT department with issues regarding e-mail, Blackboard, SelfService, on-campus labs and on-campus wi-fi, please visit https://www.hood.edu/offices-services/information-technology, or visit the help desk on the first floor of the library. You can submit a work order request from that link. For help with particular issues with the software we use in class, please ask me or your classmates.

Counseling services

If you are feeling stressed, worried, or down during the semester, or if you notice signs of emotional distress in someone else, please feel free to stop by Amanda Dymek's office (Apple Resource Building, Room 4, 301-696-3439, dymek@hood.edu) or consider reaching out for support. Hood also has many campus resources that are available to support you including:

- Hood Counseling Services – open M-F, 8:30AM-5:00PM, Apple Resource building, 1st floor, Room A. Make your appointment online at hood.edu/counseling, or drop in for a walk-in session on Mondays, Wednesdays, and Fridays between the hours of 11:00AM – 12:00PM. If you have questions about Counseling Services, email counselingservices@hood.edu
- Hood also has several connections to other mental health resources including the NeighborHood Counseling Training Center and Thriving Campus.

If you or someone you know needs to talk to someone right now, text or call 9-8-8 for a free, confidential conversation with a trained counselor 24/7

General Hood College information

Accessibility

This course is intended to be accessible for all students, including those with mental, physical, or cognitive disabilities, illnesses, injuries, impairments, or any other condition that tends to negatively affect one's equal access to education. If at any point in the term, you find yourself not able to fully access the space, content, and experience of this course, you are welcome (and not required) to contact me by email, phone, or during office hours to discuss your specific needs. I also encourage you to contact the Office of Accessibility Services (301-696-3569 or accessibilityservices@hood.edu). If you have a diagnosis or history of accommodations in high school or previous postsecondary institutions, Accessibility Services can help you document your needs and create an accommodation plan. By making a plan through Accessibility Services, you can ensure appropriate accommodations without disclosing your condition or diagnosis to course instructors. The link to the office above will take you to an online form to begin this process.

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Title IX

Hood College faculty are committed to helping create a safe and open learning environment for all students. If you (or someone you know) have experienced any form of sexual misconduct, including sexual assault, dating or domestic violence, or stalking, know that help and support are available. The College strongly encourages all members of the community to take action, seek support, and report incidents of sexual misconduct to the Title IX Coordinator. Please be aware that under Title IX of the Education Amendments of 1972, I am required to disclose information about such misconduct to the Title IX office. If you wish to speak to a confidential employee who does not have this reporting responsibility, you can contact the Dean of the Chapel (chapel@hood.edu), counseling services (counselingservices@hood.edu) , or health services (healthservices@hood.edu). For more information about reporting options and resources at Hood College and the community, please the contact the Title IX Coordinator (titleix@hood.edu) or visit https://www.hood.edu/hood-community/hood-college-health-wellness/sexual-respect-title-ix.

Hood's commitment to inclusivity and campus values

Hood College is proud of its diverse community, and we are committed to cultivating and strengthening an inclusive, tolerant, multi-cultural, and intellectually open community with equal opportunity for all. By encouraging and celebrating our differences, we create an environment that promotes freedom of thought and academic excellence. It is our goal to have a respectful and nurturing academic community that affirms the inherent worth and dignity of every individual, and celebrates the diverse backgrounds of all students, faculty, and staff. We will strive to value each person for their uniqueness and difference and to encourage all community members to reach their fullest potential.

Student Success services

Your success in my class is a priority. For this reason, I will be using the Beacon as an early identification and intervention tool. If I notice that you are struggling with issues such as attendance, class participation, or assignment/test performance, I may choose to send an Academic Alert through Beacon to connect you with appropriate campus resources. These referrals are designed to maximize your chances for success at Hood College, not as a reprimand or punishment. Please respond to any communications you may receive from me, your academic advisor, the Student Success Center, your coach, the dean's office, or other campus offices regarding your academic progress in this course.

Important Contacts

• **Assessment questions, Nathan Reese, assessment@hood.edu**
• **Campus Bookstore: https://hood.ecampus.com/**
• **Campus Security, Chief Thurmond Maynard, maynard@hood.edu**
• **Chapel, Rev. Beth O'Malley, chapel@hood.edu**
• **Counseling Services, counselingservices@hood.edu**
• **Interim Provost and Academic Judicial Council Chair, Dr. Paige Eager, eager@hood.edu**
• **Dean of Graduate School, Dr. April Boulton, boulton@hood.edu**
• **Dean of Students, Dr. Demetrius Johnson, deanofstudents@hood.edu**
• **Dean of Student Success, Camelia Rubulcada, studentsuccess@hood.edu**
• **Health Services, Amanda Dymek, healthservices@hood.edu**
• **IT Help Desk, helpdesk@hood.edu**
• **Instructional Technology (Blackboard), Jeff Welsh, welsh@hood.edu**
• **Counseling Services: counselingservices@hood.edu**
• **Accessibility Services: accessibilityservices@hood.edu**

While I intend this syllabus to be full and complete, deviations may be necessary. I will alert you to any changes if or when they occur.